



SIMATS ENGINEERING

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Course Code: CSA0619

Slot: D

Course Name: Design and Analysis of Algorithms

Course Faculty: Dr. Mary Harin Fernandez F

Project Title: Efficiency Analysis of Polynomial Algorithms Analyze an Algorithm with Complexity $O(n^3)$

Module Photographs:



Project Description: This project focuses on the Efficiency Analysis of Polynomial Algorithms with a time complexity of $O(n^3)$ using the Standard Matrix Multiplication algorithm as the model. The algorithm was analyzed to understand how its execution time increases with input size and to evaluate its scalability. A complexity comparison with $O(n)$, $O(n \log n)$, and $O(n^2)$ algorithms was performed to illustrate the growth trend. An architecture diagram, algorithm, Python implementation, complexity analysis, and graphical comparison were prepared to explain the algorithm's performance. The study concludes that although $O(n^3)$ algorithms are suitable for small and medium-sized datasets, they become inefficient for large-scale problems due to their rapidly increasing execution time.

Student Signature

Guide Signature